

Django on IIS - Complete Troubleshooting Guide

Error: HTTP Error 500.0 - Internal Server Error
Issue: FastCGI process exited unexpectedly (Error Code: 0x00000067)

Problem Analysis

Based on the error screenshots provided:

- 1. **Permissions are set correctly ✓** - IIS_IUSRS has Read & execute permissions
- 2. **Error persists** - Handler is "Django1" with error code 0x00000067
- 3. **Port changed to 8003** - Accessing localhost:8003

Error Code 0x00000067 means "The network name cannot be found" - FastCGI cannot execute the Python process or find the WSGI handler.

Root Cause

The issue is in the FastCGI configuration:

- Python executable path might be incorrect
 - The wfastcgi.py path might be wrong
 - Environment variables are not set correctly
 - Django settings module is misconfigured
-

Recommended Solution: Configure Django in IIS Manager

Step 1: Install and Enable wfastcgi

Open **Command Prompt as Administrator**:



bash

```
cd F:\irokon\project\Django-atheeb-information-management-system\Backend
env\Scripts\activate
pip install wfastcgi
wfastcgi-enable
```

Important: Copy the output path - it will look like:



```
"F:\irokon\...\env\Scripts\python.exe|F:\irokon\...\env\Lib\site-packages\wfastcgi.py"
```

Step 2: Configure FastCGI Settings in IIS Manager

1. **Open IIS Manager**
2. Click on your **server name** (top level, e.g., "DESKTOP-M5Q2LLM")
3. Double-click "**FastCGI Settings**" in the center panel
4. Click "**Add Application...**" on the right side
5. Fill in the dialog:
 - **Full Path:** F:\irokon\project\Django-atheeb-information-management-system\Backend\env\Scripts\python.exe
 - **Arguments:** F:\irokon\project\Django-atheeb-information-management-system\Backend\env\Lib\site-packages\wfastcgi.py
 - **Maximum Instances:** 4
 - **Activity Timeout:** 600
 - **Request Timeout:** 600
 - **Instance MaxRequests:** 10000
6. Click on "**Environment Variables**" (the ... button)
7. Add these environment variables:

Variable Name	Value
PYTHONPATH	F:\irokon\project\Django-atheeb-information-management-system\Backend
DJANGO_SETTINGS_MODULE	YOUR_PROJECT_NAME.settings
WSGI_HANDLER	django.core.wsgi.get_wsgi_application()
WSGI_LOG	F:\irokon\project\logs\wfastcgi.log

Note: Replace YOUR_PROJECT_NAME with your actual Django project folder name (the one containing settings.py and wsgi.py)

8. Click **OK** to save

Step 3: Configure Handler Mapping

1. In **IIS Manager**, select your website: "**Django-atheeb-information-management-system**"
2. Double-click "**Handler Mappings**"
3. Click "**Add Module Mapping...**" on the right side
4. Fill in:
 - **Request path:** *
 - **Module:** FastCgiModule (select from dropdown)
 - **Executable:**



F:\irokon\project\Django-atheeb-information-management-system\Backend\env\Scripts\python.exeF:\irokon\project\Django-atheeb-information-management-system\Backend\env\Lib\site-packages\wfastcgi.py

- **Name:** Django
5. Click "**Request Restrictions...**" button:
 - **Mapping tab:** Select "Invoke handler only if request is mapped to: File or folder"
 - **Verbs tab:** Select "All verbs"
 - **Access tab:** Select "Script"
 6. Click **OK**
 7. When prompted "Do you want to create a FastCGI application for this executable?", click **Yes**

Step 4: Set Application Pool Identity

1. In **IIS Manager**, click "**Application Pools**" on the left
 2. Find your application pool (probably same name as your site)
 3. Right-click → "**Advanced Settings...**"
 4. Under **Process Model**, find **Identity**
 5. Click "..." button
 6. Select "**ApplicationPoolIdentity**" or "**NetworkService**"
 7. Click **OK**
-

Step 5: Create Logs Directory

Create this folder:



F:\irokon\project\logs\

Set permissions:

1. Right-click folder → Properties → Security → Edit
 2. Add **IIS_IUSRS** with **Modify** permissions
 3. Add **IUSR** with **Modify** permissions
 4. Click **OK**
-

Step 6: Configure Error Pages (Temporarily)

1. Select your website in **IIS Manager**
 2. Double-click "**Error Pages**"
 3. Click "**Edit Feature Settings...**" on the right
 4. Select "**Detailed errors**"
 5. Click **OK**
-

Step 7: Test Django Standalone First

Before testing in IIS, verify Django works:



bash

```
cd F:\irokon\project\Django-atheeb-information-management-system\Backend
env\Scripts\activate
python manage.py check
python manage.py runserver 0.0.0.0:8003
```

Open browser → http://localhost:8003

If this fails, fix Django first before continuing with IIS.

Step 8: Restart IIS

Option 1 - Using IIS Manager:

- 1. Right-click your **server name**
- 2. Click "**Stop**"
- 3. Wait 5 seconds
- 4. Right-click again → "**Start**"

Option 2 - Using Command Prompt (as Administrator):



bash

```
iisreset
```

Step 9: Check Logs

After accessing your site, check the log file:



```
F:\irokon\project\logs\wfastcgi.log
```

This will show the **actual Django error** and help identify the issue.

Common Issues Checklist

Issue 1: Wrong Django Settings Module

- Find your Django project folder (contains `settings.py` and `wsgi.py`)
- Use that folder name in `DJANGO_SETTINGS_MODULE`
- Example: If structure is `Backend/myproject/settings.py`, use `myproject.settings`

Issue 2: Static Files Configuration

In your Django `settings.py`, temporarily set:



python

```
ALLOWED_HOSTS = ['*']  
DEBUG = True # for testing only
```

Issue 3: Database Permissions

- Ensure SQLite file (if using) has write permissions for IIS_IUSRS
- Verify database connection settings are correct

Alternative Solution: Reverse Proxy (Recommended)

If FastCGI continues to fail, use this **more reliable approach**:

Why Use Reverse Proxy?

-  **Much easier to debug** - Django runs separately
-  **More reliable** - FastCGI on Windows is problematic
-  **Better performance** - Waitress is optimized for Windows
-  **Easier to restart** - just restart Waitress, not IIS
-  **Better logging** - Django logs work normally

Implementation Steps

1. Install Required IIS Modules

URL Rewrite Module:

- Download from: <https://www.iis.net/downloads/microsoft/url-rewrite>
- Install it

Application Request Routing (ARR):

- Download from: <https://www.iis.net/downloads/microsoft/application-request-routing>
- Install it
- Open IIS Manager → Server level → "Application Request Routing"
- Click "Server Proxy Settings" → Check "Enable proxy" → Apply

2. Install Waitress



bash

```
cd F:\irokon\project\Django-atheeb-information-management-system\Backend
env\Scripts\activate
pip install waitress
```

3. Create Startup Script

Create file: F:\irokon\project\start_django.bat



batch

@echo off

cd F:\irokon\project\Django-atheeb-information-management-system\Backend

call env\Scripts\activate

waitress-serve --listen=127.0.0.1:8003 YOUR_PROJECT_NAME.wsgi:application

Replace YOUR_PROJECT_NAME with your Django project folder name.

4. Test Django with Waitress

Run the script:



bash

start_django.bat

Test in browser: <http://localhost:8003>

5. Configure URL Rewrite in IIS Manager

1. Select your website in IIS Manager
2. Double-click "**URL Rewrite**"
3. Click "**Add Rule(s)...**" on the right
4. Select "**Reverse Proxy**"
5. If prompted to enable proxy, click **Yes**
6. Enter server name: 127.0.0.1:8003
7. Click **OK**

6. Create Windows Service (For Production)

Use **NSSM** (Non-Sucking Service Manager) to run Django as a service:

1. Download NSSM from: <https://nssm.cc/download>
2. Extract to a folder (e.g., C:\nssm\)
3. Open Command Prompt as Administrator:



bash

cd C:\nssm\win64

nssm install DjangoService "F:\irokon\project\start_django.bat"

nssm set DjangoService AppDirectory "F:\irokon\project"

nssm set DjangoService Description "Django Application Service"

nssm start DjangoService

The service will now start automatically with Windows.

Comparison: FastCGI vs Reverse Proxy

Feature	FastCGI	Reverse Proxy
Setup Complexity	High	Medium
Debugging	Difficult	Easy
Reliability on Windows	Poor	Excellent
Performance	Good	Excellent
Restart Process	Restart IIS	Restart Service
Log Visibility	Limited	Full Django logs
Recommendation	⚠️ Not Recommended	✅ Recommended

Deployment Package Structure

For client deployment, package as follows:



```
deployment-package/
├── frontend/
│   └── dist/           # Built Angular app
├── backend/
│   ├── django-app/    # Your Django code
│   ├── packages/      # Downloaded pip packages
│   ├── python-embedded/ # Python runtime
│   └── requirements.txt
├── tools/
│   ├── nssm/          # Service manager
│   └── installers/     # IIS modules
├── scripts/
│   ├── install.ps1    # Main installation script
│   ├── start_django.bat # Django startup script
│   ├── setup_service.ps1 # Service configuration
│   └── configure_iis.ps1 # IIS setup script
└── documentation/
    ├── installation-guide.pdf
    ├── configuration-guide.pdf
    └── troubleshooting-guide.pdf
```

Installation Script for Client

Create `install.ps1`:



powershell

Check if running as Administrator

```
if (-NOT ([Security.Principal.WindowsPrincipal][Security.Principal.WindowsIdentity]::GetCurrent()).IsInRole([Security.Principal.WindowsBuiltInRole] "Administrator")) {  
    Write-Warning "Please run this script as Administrator!"  
    exit  
}
```

Write-Host "Installing Django Application..." -ForegroundColor Green

Install Python dependencies offline

Write-Host "Installing Python packages..." -ForegroundColor Yellow

cd backend

python-embedded\python.exe -m pip install --no-index --find-links=packages -r requirements.txt

Run Django migrations

Write-Host "Running database migrations..." -ForegroundColor Yellow

python-embedded\python.exe manage.py migrate

Collect static files

Write-Host "Collecting static files..." -ForegroundColor Yellow

python-embedded\python.exe manage.py collectstatic --noinput

Install Windows Service

Write-Host "Installing Django service..." -ForegroundColor Yellow

cd ..\tools\nssm\win64

.\nssm.exe install DjangoService "\$PWD\..\..\scripts\start_django.bat"

.\nssm.exe start DjangoService

Configure IIS

Write-Host "Configuring IIS..." -ForegroundColor Yellow

Copy frontend files to IIS directory

Configure URL Rewrite rules

Write-Host "Installation completed successfully!" -ForegroundColor Green

Write-Host "Please configure IIS URL Rewrite manually as described in documentation" -ForegroundColor Yellow

Information Needed for Exact Configuration

To provide the **exact configuration values**, please provide:

1. **Django project name** - The folder containing `settings.py` and `wsgi.py`
 - Look in the Backend folder structure
2. **Django standalone test result** - Run this and share the output:



bash

```
cd F:\irokon\project\Django-atheeb-information-management-system\Backend
env\Scripts\activate
python manage.py check
python manage.py runserver
```

3. **Log file contents** - After attempting to access via IIS, share:



```
F:\irokon\project\logs\wfastcgi.log
```

Support Resources

- **Django Documentation:** <https://docs.djangoproject.com/>
- **IIS Documentation:** <https://docs.microsoft.com/en-us/iis/>
- **wfastcgi GitHub:** <https://github.com/microsoft/wfastcgi>
- **Waitress Documentation:** <https://docs.pylonsproject.org/projects/waitress/>
- **NSSM Documentation:** <https://nssm.cc/usage>

Conclusion

Recommended Approach:

1. **First Choice:** Use the **Reverse Proxy with Waitress** approach
 - More reliable and easier to debug
 - Better suited for Windows/IIS environment
 - Easier for client to maintain
2. **Second Choice:** FastCGI configuration
 - Only if reverse proxy is not feasible
 - Requires more careful configuration
 - More difficult to troubleshoot

For production deployment to clients without internet, package everything including Python runtime, all dependencies, and automated installation scripts.

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